



NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA-769008
END-SEMESTER EXAMINATION, 2024
SESSION: 2023 – 2024 (Spring)
 B. Tech. 8th / M. Tech. 2nd Semester

Subject code: CS6404 **Subject Name:** Software Testing
No. of pages: 3 **Full Marks:** 100

Dept. Code: CS
Duration: 3 Hours

Figures at the right-hand margin indicate marks.
All parts of a question should be answered at one place.
Answer any FIVE questions including Q1. Justify your answers.

Q. No.	Particulars	Marks
1		2*10=20
(a)	What will be the total number of test sessions required using pair-wise integration testing, if a call graph has total 20 nodes and 19 edges? Justify your answer.	
(b)	If a test suite T contains 100 tests of which 40 are modification-revealing for program P and modified program P', and the technique M selects 24 of these 40 tests, what is the inclusiveness of M relative to P, P' and T?	
(c)	Suppose in a design hierarchy, there are 20 numbers of modules / nodes, out of which 10 are, leave nodes. How many numbers of stubs will be required for the subordinate modules, in top down integration testing?	
(d)	What do you mean by relevant slice? Why it is so important in software testing?	
(e)	What is the implication of dynamic binding in testing of object-oriented software?	
(f)	Explain 2 coverage criteria used testing of O-O software using sequence diagrams.	
(g)	For class level testing, which test design pattern, will be most suitable?	
(h)	Mention the names of two tools that are used for testing of agile software.	
(i)	How do web-based applications differ from web-enabled applications?	
(j)	Differentiate between the two types of regression testing in Agile environment?	
2	Perform different types of performance testing for the software that implements the student admission process of NIT Rourkela. You may assume the suitable functionalities of your own.	12+08=20
(a)	Perform bottom up integration procedure for the following system hierarchy. Show all the steps.	
	<pre> graph TD 1[1] --> 2a[2a] 1 --> 2b[2b] 2a --> 3[3] 2a --> 4[4] 2a --> 5[5] 2b --> 6[6] 2b --> 7[7] 4 --> 8[8] 4 --> 9[9] 5 --> 10[10] 6 --> 11[11] 7 --> 12[12] 7 --> 13[13] </pre>	

3. (a)	Suppose you have to develop a program for admitting students in a Data Science course in NIT Rourkela, subject to the following conditions: (i) Marks in Python ≥ 70 (ii) Marks in Java ≥ 60 (iii) Marks in C++ ≥ 60 (iv) Total in all three subjects ≥ 220 or Total in Java and Python ≥ 150 If the aggregate mark of an eligible candidate is more than 240, he/she will be eligible for the scholarship course; otherwise he/she will be eligible for a normal course. The program reads the name of the student and marks in the three subjects and generates the following outputs: (i) Not eligible (ii) Eligible for scholarship course (iii) Eligible for normal course Design at least 5 test cases for this program using equivalence class partition testing. (b) Consider the following function: <pre>int findminimum(int i, int j, int k) { int min; if (i < j) { if (i < k) { min = i;} else { min = k;}} else { if (j < k) { min = j;} else { min = k;}} return min; }</pre> (i) Draw the CFG for the above function. (ii) Calculate cyclomatic complexity. (iii) List all the linearly independent paths. (iv) Design the test cases from the linearly independent paths.	10+10=20																																
4. (a)	Consider a Banking Information System with the following functionalities. <table><tr><th>Serial Number</th><th>Functionality</th><th>Function ID in SRS</th><th>Test Cases</th></tr><tr><td>1</td><td>Login the system</td><td>F 3.4</td><td>T1</td></tr><tr><td>2</td><td>Check balance</td><td>F 3.5</td><td>T2</td></tr><tr><td>3</td><td>Deposit money</td><td>F 3.6</td><td>T3</td></tr><tr><td>4</td><td>Withdraw money</td><td>F 3.7</td><td>T4</td></tr><tr><td>5</td><td>Transfer money</td><td>F 3.8</td><td>T5</td></tr><tr><td>6</td><td>Update KYC</td><td>F 3.9</td><td>T6</td></tr><tr><td>7</td><td>Exit</td><td>F 3.10</td><td>T7</td></tr></table> Develop the test specifications and test incidence report for Withdraw money functionality.	Serial Number	Functionality	Function ID in SRS	Test Cases	1	Login the system	F 3.4	T1	2	Check balance	F 3.5	T2	3	Deposit money	F 3.6	T3	4	Withdraw money	F 3.7	T4	5	Transfer money	F 3.8	T5	6	Update KYC	F 3.9	T6	7	Exit	F 3.10	T7	12+08=20
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(b)	Briefly explain use-case based testing with a suitable example. Mention the generic test requirements for use-case based testing.																																	

5. (a) (b)	Briefly explain the test procedure using Extended Use Case test pattern. Apply the above test procedure to the <i>Cash Replenishment</i> use case of an ATM system, and develop the test cases in a tabular manner. Briefly explain behavioral integration testing. Draw the state chart diagram for a <i>Credit Card</i> object. Generate at least 4 behavioral test cases for <i>Credit Card</i> using the state chart diagram.	10+10=20																																																																																																																									
6. (a) (b)	Why regression test prioritization is needed? Briefly explain regression test prioritization technique along with a suitable diagram and all the necessary steps. Consider a program with 10 faults and a test suite of 10 test cases, as shown in the table given below: <table><tr><td></td><td>T1</td><td>T2</td><td>T3</td><td>T4</td><td>T5</td><td>T6</td><td>T7</td><td>T8</td><td>T9</td><td>T10</td></tr><tr><td>F1</td><td></td><td></td><td></td><td></td><td>X</td><td></td><td></td><td>X</td><td></td><td></td></tr><tr><td>F2</td><td></td><td>X</td><td>X</td><td>X</td><td></td><td>X</td><td></td><td></td><td></td><td></td></tr><tr><td>F3</td><td>X</td><td>X</td><td>X</td><td>X</td><td></td><td></td><td>X</td><td>X</td><td></td><td></td></tr><tr><td>F4</td><td></td><td></td><td></td><td></td><td></td><td>X</td><td>X</td><td></td><td></td><td></td></tr><tr><td>F5</td><td></td><td></td><td>X</td><td></td><td></td><td>X</td><td></td><td></td><td></td><td>X</td></tr><tr><td>F6</td><td></td><td></td><td></td><td></td><td>X</td><td>X</td><td>X</td><td></td><td>X</td><td></td></tr><tr><td>F7</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>X</td><td></td></tr><tr><td>F8</td><td></td><td>X</td><td></td><td>X</td><td></td><td>X</td><td></td><td></td><td>X</td><td>X</td></tr><tr><td>F9</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>X</td></tr><tr><td>F10</td><td></td><td></td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td>X</td><td>X</td></tr></table> Calculate its APFD with the following orders of test cases. (i) TS1 = (T1, T2, T3, T4, T5, T6, T7, T8, T9, T10) (ii) TS2 = (T2, T1, T3, T4, T5, T6, T7, T8, T9, T10) (iii) TS3 = (T1, T2, T5, T4, T3, T6, T9, T8, T7, T10) What is the effect of changing the order of test cases? Which one is the best test suite in the context of detecting faults?		T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	F1					X			X			F2		X	X	X		X					F3	X	X	X	X			X	X			F4						X	X				F5			X			X				X	F6					X	X	X		X		F7									X		F8		X		X		X			X	X	F9	X									X	F10			X	X					X	X	10+10=20
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7. (a) (b) (c)	Briefly discuss unified process as an agile software development framework. With a suitable figure, briefly explain how testing is carried out in an agile software development framework. You are asked to develop a website for an inventory management system having the following features: (i) Buttons and boxes (ii) Field validation (iii) Static and dynamic contents (iv) Database functionalities (v) YouTube links and graphical contents Briefly explain the following types of web testing that you would perform for the above web application. a. Content testing b. Usability testing	5+7+8=20																																																																																																																									